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Technical Note

Comparative study of chromium biosorption by red, green and brown seaweed biomass

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Abstract

Dried biomass of the macroalgae *Fucus vesiculosus* and *Fucus spiralis* (brown), *Ulva* spp. (comprising *Ulva linza*, *Ulva compressa* and *Ulva intestinalis*) and *Ulva lactuca* (green), *Palmaria palmata* and *Polysiphonia lanosa* (red) were studied in terms of their chromium biosorption performance. Metal sorption was highly pH dependent with maximum Cr(III) and Cr(VI) sorption occurring at pH 4.5 and pH 2, respectively. Extended equilibrium times were required for Cr(VI) binding over Cr(III) binding (180 and 120 min, respectively) thus indicating possible disparities in binding mechanism between chromium oxidation states. The red seaweed *P. palmata* revealed the highest removal efficiency for both Cr(III) and Cr(VI) at low initial concentrations. However, at high initial metal concentrations *F. vesiculosus* had the greatest removal efficiency for Cr(III) and performed almost identically to *P. lanosa* in terms of Cr(VI) removal. The Langmuir Isotherm mathematically described chromium binding to the seaweeds where *F. vesiculosus* had the largest q_{max} for Cr(III) sorption (1.21 mmol g^{-1}) and *P. lanosa* had the largest Cr(VI) uptake (0.88 mmol g^{-1}). *P. palmata* had the highest affinity for both Cr(III) and Cr(VI) binding with *b* values of 4.94 mM^{-1} and 8.64 mM^{-1}, respectively. Fourier transform infrared analysis revealed interactions of amino, carboxyl, sulphonate and hydroxyl groups in chromium binding to *Ulva* spp. The remaining seaweeds showed involvement of these groups to varying degrees as well as ether group participation in the brown seaweeds and for Cr(VI) binding to the red seaweeds.
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1. Introduction

Heavy metal pollution is one of the most significant problems of this century (Park et al., 2006). Chromium has found widespread use in electroplating, leather tanning and metal finishing with Cr(VI) being a suspected carcinogen and mutagen (Costa, 2003). Thus, the discharge of Cr(VI) into surface water is regulated to below 0.05 mg l^{-1} by the U.S. EPA (Baral and Engelken, 2002). Conventional chromium removal processes including ion-exchange, activated carbon adsorption, reverse osmosis, and membrane filtration can be expensive or ineffective at low concentrations and may also lead to secondary environmental problems from waste disposal (Arslan and Pehlivan, 2007). Thus, much research has been focussed on identifying biological materials that can efficiently remove heavy metals from aqueous environments. These materials are known as biosorbents and the passive binding of metals by living or dead biomass is referred to as biosorption (Schiewer and Wong, 2000).

Seaweeds are extremely efficient biosorbents with the ability to bind various metals from aqueous effluents (Davis et al., 2003; Tsui et al., 2006). Numerous chemical groups may be responsible for metal biosorption by seaweeds e.g. carboxyl, sulphonate, hydroxyl and amino (Smith and Lacher, 2002) with their relative importance depending on factors such as the quantity of sites, their accessibility and the affinity between site and metal. The

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main metal binding mechanisms include ion-exchange and complex formation (Davis et al., 2003) but these may differ according to biomass type, origin and the processing to which it has been subjected.

This paper studies chromium binding to seaweeds from each of the three main classes available off the South-East coast of Ireland. The seaweeds under investigation are *Fucus vesiculosus* and *Fucus spiralis* (brown), *Ulva* spp. and *Ulva lactuca* (green), *Palmaria palmata* and *Polysiphonia lanosa* (red). Cr(III) and Cr(VI) binding to dead seaweed biomass is described in terms of binding capacity and affinity while the effects of manipulation of experimental parameters e.g. pH, equilibrium time and initial metal concentration on binding are also investigated. Fourier transform infrared spectroscopy (FTIR) is also employed to identify the specific functionalities involved in chromium binding to these seaweeds.

2. Materials and methods

2.1. Biomass

Seaweeds were harvested from Fethard-on-Sea, Co. Wexford, Ireland (52°11′53.68″N, 6°49′34.36″W), rinsed thoroughly with distilled water, oven-dried at 60 °C for 24 h, then subsequently ground and sieved to a particle size of 500–850 μm (Raw biomass).

2.2. Metal solutions

Working solutions were prepared by dilution of chromium stock solutions (19 mM) with distilled water. Solution pH was adjusted using 0.1 M NaOH and HCl as required. Initial and equilibrium metal concentrations were subsequently determined using AAS (SpectrAA-600 VARIAN, Software version 4.10, flame mode).

2.3. Optimum pH determination

One hundred milligrams of raw biomass were added to 50 ml of 1 mM Cr(III) and Cr(VI) solutions adjusted to pH 1.5–6. Flasks were shaken for 6 h at 200 rpm and room temperature (RT) (21 ± 1 °C). Samples were filtered under vacuum and the filtrate was analysed via AAS. The equilibrium metal uptake q (mmol g^{-1}) was calculated according to an equation used by Lodeiro et al. (2004).

2.4. Determination of equilibrium time

One hundred milligrams of raw biomass were added to 50 ml of 1 mM Cr(III) and Cr(VI) solutions adjusted to pH 4.5 and pH 2, respectively. Flasks were shaken at 200 rpm and RT with samples removed at intervals of 5, 10, 30, 60, 120, 240, 300 and 360 min and subsequently analysed by AAS. The metal uptake at time t (q_t) was calculated according to an equation adapted from Lodeiro et al. (2004).

2.5. Batch adsorption experiments

One hundred milligrams of raw biomass were added to 50 ml of Cr(III) and Cr(VI) solutions of various concentrations (0.09–4.8 mM) adjusted to pH 4.5 and pH 2, respectively. Flasks were shaken for 4 h at RT and 200 rpm. After suitable dilution, metal uptake was measured using AAS and q was then calculated. Data was analysed using the linearised Langmuir equation which yielded q_{max}, the maximum metal uptake (mmol g^{-1}) and b, an affinity parameter (mM^{-1}).

2.6. Fourier transform infrared spectroscopy

Protonated biomass was used as the biosorbent control during FTIR analysis and was prepared according to a procedure adapted from Fourest and Volesky (1996). Protonated biomass (at a concentration of 2 mg ml^{-1}) was exposed to 4 μM Cr(III) and Cr(VI) solutions (adjusted to pH 4.5 and 2, respectively) over a 4 h period. Samples were filtered under vacuum, oven-dried at 60 °C for 24 h and analysed directly using a Digilab Scimitar Series infrared spectrometer (MIRacle™ Single Reflection HATR diamond accessory). A background scan with the diamond in place was run before each analysis. Triplicate samples were analysed over 40 scans at a resolution of 2 cm^{-1}.

3. Results and discussion

3.1. Optimum pH determination

Chromium-containing wastewaters are usually acidic and thus, pH experiments in this study were conducted in the pH range 1.5–6. Various authors (Figueira et al., 1999; Chen et al., 2002; Davis et al., 2003) have shown that solution pH greatly influences metal biosorption by seaweeds. Functional groups such as sulphonate (–OSO_3) and carboxyl (–COOH) display acidic characteristics and therefore, the pH at which maximum metal uptake occurs is related to the pK_a of these groups. The point of zero charge (PZC) of the seaweeds was determined from potentiometric titrations in a previous study by Murphy et al. (2007). *F. vesiculosus* and *F. spiralis* were shown to have PZC values of 6.05 and 5.85, respectively, *U. lactuca* and *Ulva* spp. had values of 6.45 and 6.13 while the values for *P. palmata* and *P. lanosa* were 6.18 and 6.41. These results are comparable with those reported by Romero-Gonzalez et al. (2001) for dealginated seaweed waste where the PZC was 6.36. Therefore, below the PZC, the biomass still has a net positive charge despite the presence of dissociated negatively charged functionalities.

Cr(III) is cationic in solution (Cr^{3+} and $CrOH^{2+}$) while Cr(VI) occurs as anionic species such as CrO_4^{2-} and $HCrO_4^-$. Therefore, changes in pH should have significantly different effects on the binding behaviour of the chromium species. The relationship between chromium uptake and solution pH is illustrated in Fig. 1.

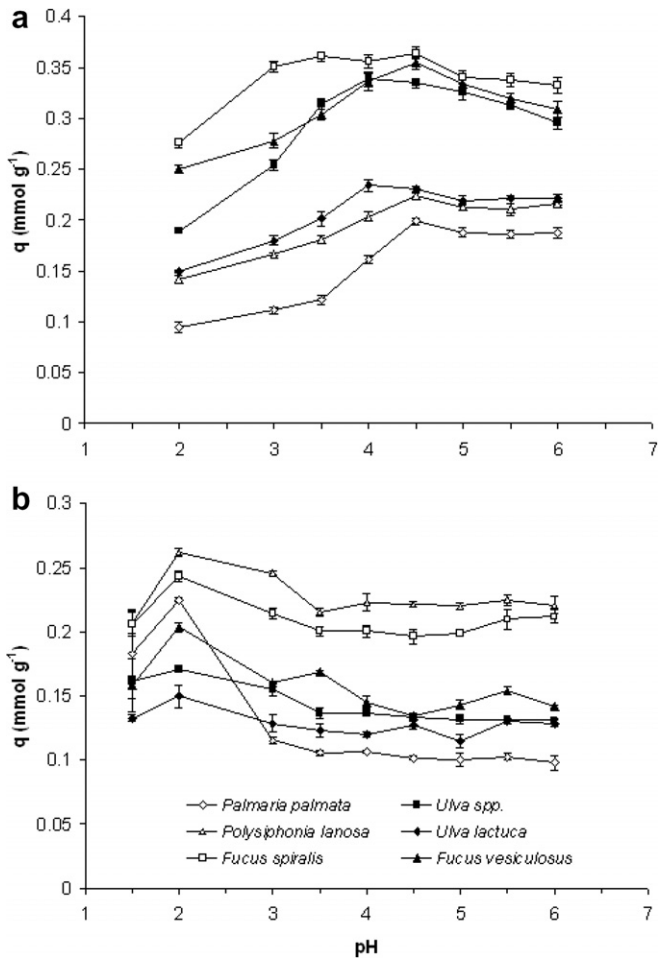


Fig. 1. Determination of the optimum pH for (a) Cr(III) and (b) Cr(IV) sorption. Error bars are calculated based on triplicate runs with 95% confidence intervals. Initial metal concentration = 1 mM, biomass concentration = 2 mg ml^{-1}.

It is clearly seen that, in Fig. 1a, up to pH 4.5, an increase in pH resulted in increased Cr(III) uptake. At pH values <3, competition from H^+ ions in solution for available biomass binding sites contributes to decreased Cr(III) binding. However, as pH increases, fewer H^+ ions are present in solution resulting in less competition for binding sites and hence increasing Cr(III) uptake. The pH dependence of Cr(III) uptake suggests that carboxyl groups (pK_a 3.5–5.0) are important in binding (Sheng et al., 2004). Murphy et al. (2007) identified various carboxyl functionalities with pK_a values in this range for the seaweeds under investigation e.g. *F. vesiculosus* (3.85 ± 0.1 and 4.68 ± 0.2) and *Ulva* spp. (3.69 ± 0.3 and 4.59 ± 0.3). Therefore, it was expected that carboxyl groups would significantly influence metal binding in these seaweeds. Sheng et al. (2004) also showed that sulphonate groups (pK_a 1.0–2.5) may contribute to metal binding at low pH. The interaction of these functionalities with Cr(III) is evident in Fig. 1a, where there is decreased, but not negligible metal uptake at low pH.

At pH >3.5, dissociation of sulphonate and carboxyl functionalities leads to increased negative charge on the biomass, so, while anionic Cr(VI) species are repelled, cationic Cr(III) experiences an increased attraction to the biomass resulting in improved metal binding.

From Fig. 1a, pH 4.5 was chosen as the optimum pH for Cr(III) sorption. At this pH, maximum uptake was achieved and metal precipitation as $Cr(OH)_3$ was avoided (Yun et al., 2001). Haug and Smidsrod (1970) showed that, in the pH range 1–5, Cr^{3+} and $CrOH^{2+}$ are the major species present in solution with the affinity of Cr^{3+} for biomass binding sites being significantly larger than that of $CrOH^{2+}$. At pH values <3.5, $CrOH^{2+}$ binding remains at a level lower than that of Cr^{3+} but gradually increases with increasing pH and eventually exceeds the level of Cr^{3+} binding at pH >4.5, possibly explaining the decreased Cr(III) binding observed above this pH.

The optimum pH for Cr(VI) sorption was pH 2 (Fig. 1b). Increased solution pH increases the negative charge on the biomass thus repelling anionic Cr(VI) species. As solution pH decreases, amino and carboxyl groups may become protonated, thus making the biomass more positively charged and hence creating an electrostatic attraction with Cr(VI) species.

3.2. Determination of equilibrium time for Cr(III) and Cr(VI) sorption

Fig. 2 illustrates the equilibrium time required for chromium uptake by the six seaweeds. Both Cr(III) and Cr(VI) binding resulted in similar kinetic patterns with rapid initial sorption followed by a long period of equilibrium. After this equilibrium period, metal uptake did not change considerably with time. This result is in agreement with those found for a number of similar biosorbent types (Aksu, 2002; Sheng et al., 2004).

The equilibrium time required for different biomass–metal systems in this study ranged from 30 to 180 min with Cr(III) binding requiring shorter equilibrium times in all cases. The red seaweeds each required 30 min for Cr(III) uptake, *U. lactuca* required 60 min while, *Ulva* spp, *F. vesiculosus* and *F. spiralis* each took 120 min to reach equilibrium. Conversely, Cr(VI) equilibrium required 120 min for *Ulva* spp., *U. lactuca* and *P. palmata* with the remaining seaweeds reaching equilibrium within 180 min of exposure. Disparities in equilibrium time between Cr(III) and Cr(VI) point to obvious mechanistic differences in binding but this finding requires further investigation which is outside the immediate scope of this paper.

3.3. Adsorption isotherms

Fig. 3 illustrates the experimental isotherm plots obtained for Cr(III) and Cr(VI) binding. Using the data in Fig. 3, removal efficiencies (RE) at high and low initial metal concentrations (C_i) were calculated (Table 1).

At low C_i (0.09 mM), *P. palmata* displayed the greatest RE for Cr(III) and Cr(VI). This value was significantly decreased at high C_i (3.84 mM) thus indicating the

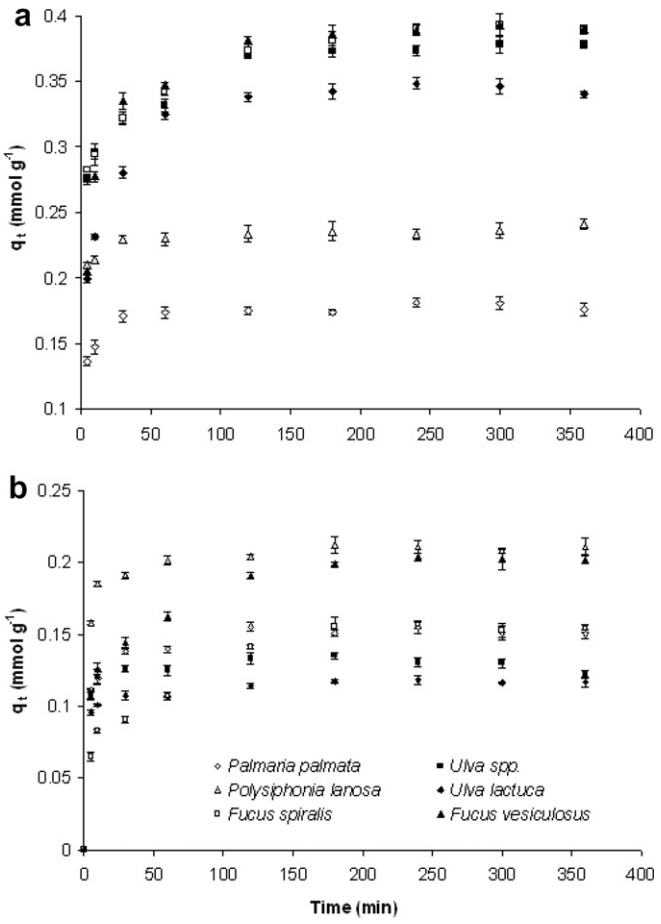


Fig. 2. Determination of equilibrium binding time for (a) Cr(III) and (b) Cr(VI) sorption. Error bars are calculated based on triplicate runs with 95% confidence intervals. Initial metal concentration = 1 mM, biomass concentration = 2 mg ml^{-1}, pH = 4.5 and 2, respectively.

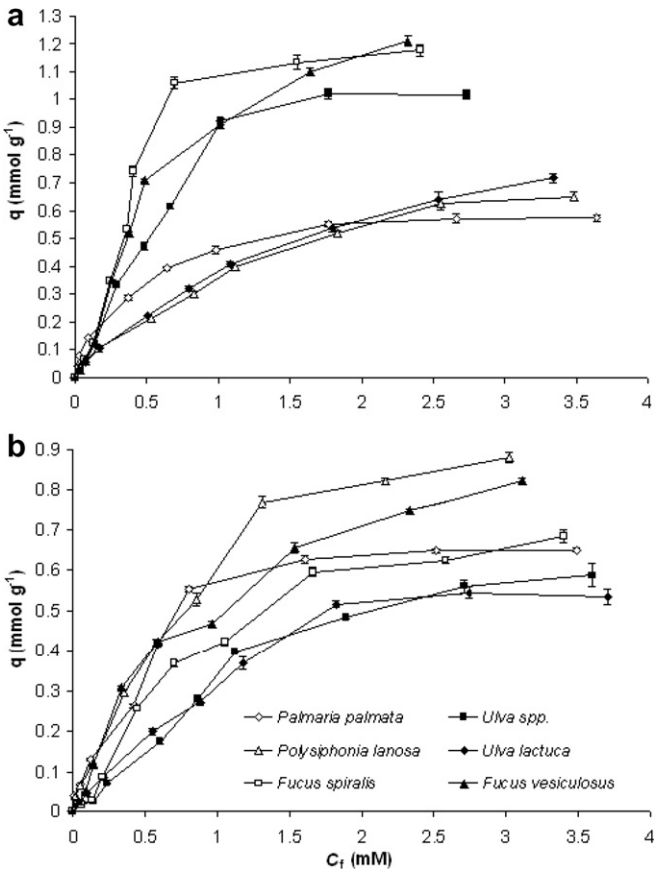


Fig. 3. Experimental isotherm data for (a) Cr(III) and (b) Cr(VI) sorption. Biomass concentration = 2 mg ml^{-1}. pH = 4.5 and 2, respectively. Error bars are calculated based on triplicate runs with 95% confidence intervals.

suitability of this seaweed for treatment of dilute metal solutions. On the other hand, at high C_i, *F. vesiculosus* displayed the highest RE for Cr(III), but for Cr(VI), both *F. vesiculosus* and *P. lanosa* had the same RE (within error) indicating their potential applicability for treatment of more concentrated metal solutions. RE values reflect the increased number of metal ions competing for available binding sites at high C_i. At low C_i, the number of available binding sites on the biomass is high (relative to the metal concentration) and hence metal uptake is very efficient.

The Langmuir isotherm was also used to model experimental isotherm data in Fig. 3. Langmuir parameters provide useful information regarding the sorption process in terms of maximum uptake capacity (q_{max}) and affinity (b). Although q_{max} is dependent on experimental conditions such as solution pH and temperature, it is a good measure for comparing different sorbents for the same metal.

The initial isotherm gradient indicates the sorbent affinity at low metal concentrations. In the Langmuir equation, this initial gradient corresponds to the affinity constant b. Davis et al. (2003) showed that a high b value indicates a steep desirable beginning of the isotherm which reflects high affinity of the sorbent for the metal.

Table 1
Removal efficiencies for chromium binding to six seaweeds

	Cr(III) (%) C_i (mM)		Cr(VI) (%) C_i (mM)	
	0.09	3.84	0.09	3.84
Fucus vesiculosus	74 ± 1	50 ± 2	76 ± 3	39 ± 1
Fucus spiralis	70 ± 3	43 ± 2	56 ± 3	35 ± 1
Ulva lactuca	73 ± 1	30 ± 1	50 ± 3	23 ± 1
Ulva spp.	71 ± 1	43 ± 2	45 ± 1	25 ± 1
Palmaria palmata	87 ± 1	24 ± 1	91 ± 2	28 ± 1
Polysiphonia lanosa	63 ± 2	27 ± 1	81 ± 2	37 ± 2

Errors are calculated based on triplicate analyses with 95% confidence intervals.

Plots of $1/q$ vs. $1/C_f$ were constructed according to the linearised Langmuir equation resulting in a straight line with a slope of $(1/(b \cdot q_{max}))$ and an intercept of $(1/q_{max})$. Langmuir parameters q_{max} and b are summarised in Table 2.

The q_{max} values obtained for Cr(III) binding decreased in the order: brown > green > red with *F. vesiculosus* and *P. palmata* having the highest and lowest values, respectively. The q_{max} values obtained by Tsui et al. (2006) for Cr(III) binding to *Sargassum hemiphyllum* (1.39 mmol g^{-1})

Table 2
Langmuir parameters for chromium biosorption by six seaweeds

	Cr(III)			Cr(VI)		
	q_{max} (mmol g^{-1})	b (mM^{-1})	r^2	q_{max} (mmol g^{-1})	b (mM^{-1})	r^2
Fucus vesiculosus	1.21 ± 0.12	1.88 ± 0.20	0.98	0.82 ± 0.04	1.76 ± 0.16	0.98
Fucus spiralis	1.17 ± 0.09	1.77 ± 0.13	0.99	0.68 ± 0.05	1.47 ± 0.15	0.97
Ulva lactuca	0.71 ± 0.04	1.98 ± 0.19	0.94	0.53 ± 0.07	1.97 ± 0.19	0.99
Ulva spp.	1.02 ± 0.10	1.38 ± 0.12	0.99	0.58 ± 0.07	1.20 ± 0.12	0.97
Palmaria palmata	0.57 ± 0.03	4.94 ± 0.34	0.98	0.65 ± 0.09	8.64 ± 0.31	0.86
Polysiphonia lanosa	0.65 ± 0.03	1.34 ± 0.11	0.99	0.88 ± 0.11	2.44 ± 0.23	0.94

Errors are calculated based on triplicate analyses with 95% confidence intervals.

are well matched with those found for the brown seaweeds in this study while a q_{max} of 0.54 mmol g^{-1} for *Spirogyra* biomass (Bishnoi et al., 2007) was also comparable with the results obtained for *P. palmata* in this study.

Although q_{max} indicates overall capacity, biomass performance is likely to be determined by both q_{max} and b values. For example, a biosorbent with a low q_{max} and a high b could outperform a biosorbent with a high q_{max} and a low b, especially in cases where metal ions are present in trace amounts (Hashim and Chu, 2004). In this study, despite *P. palmata* having a q_{max} of less than half that of *F. vesiculosus*, it possessed the largest b value thus indicating its high affinity for Cr(III) at low equilibrium concentrations. Therefore, the most suitable seaweed for Cr(III) sorption depends on its final application with *F. vesiculosus* being most suitable for high concentration applications and *P. palmata* being most suitable for low residual concentrations. In the case of *P. palmata*, its potential suitability as a biosorbent for Cr(III) is not readily apparent if the seaweeds are compared purely on the basis of q_{max}.

In contrast to Cr(III), isotherm analysis of Cr(VI) binding (Table 2) revealed that the red seaweed *P. lanosa* had the greatest q_{max}. However, similarly to Cr(III) binding, the affinity parameter, b, was highest for Cr(VI) binding to *P. palmata*. Therefore, *P. palmata* has high affinity for both cations and anions at low residual concentrations. From the values obtained in this study, it appears that the most suitable seaweed for Cr(VI) biosorption is *P. lanosa* as it possessed comparatively large q_{max} and b values. This result is in agreement with those found by various authors detailing the improved sorption performance of red seaweeds for Cr(VI) (Nourbakhsh et al., 1994; Cabatingan et al., 2001).

3.4. Fourier transform infrared spectroscopy (FTIR)

FTIR spectroscopy has previously been used to detect vibrational frequency changes in seaweeds and allows identification of the functionalities capable of interacting with metal ions (Figueira et al., 1999; Park et al., 2004; Sheng et al., 2004). Assignment of the FTIR peaks to specific functional groups is based on the work of Clothup et al. (1990). FTIR spectra of protonated, Cr(III) and Cr(VI) loaded *Ulva* spp. are shown in Fig. 4 and detail the significant band shifts observed after chromium binding to the seaweed.

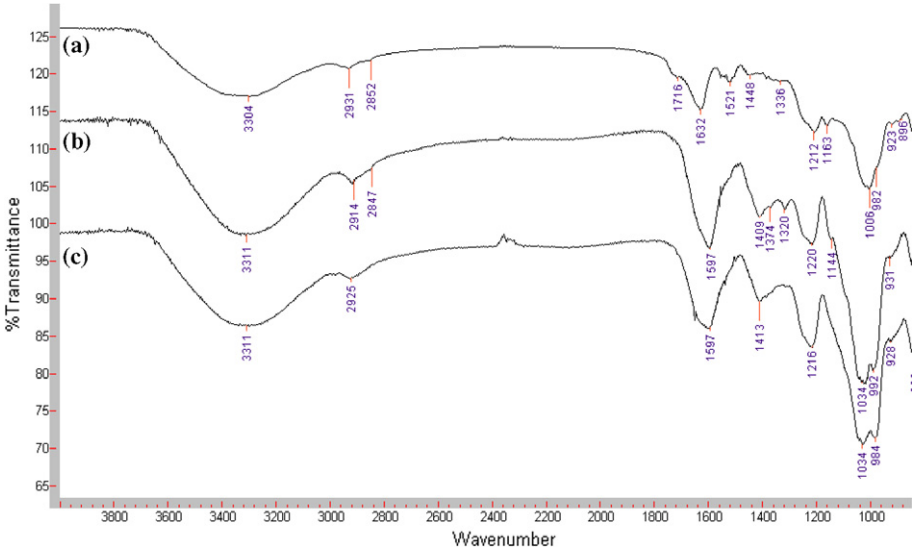


Fig. 4. Stretching frequencies observed for (a) protonated and (b) Cr(III)-loaded (c) Cr(VI)-loaded *Ulva* spp. Number of scans = 40, resolution = 2. Sample spectra from triplicate analyses are shown.

Considerable amino interaction was evident from the disappearance of the –NH stretching band (1525 cm^{-1}) in the Cr(III) and Cr(VI)-loaded spectra (Fig. 4b and c). "Free" carboxyl groups gave rise to a stretching band at 1716 cm^{-1} (Fig. 4a) which was not present after chromium exposure indicating participation of these functionalities in metal binding. A strong asymmetric C=O stretch initially present at 1632 cm^{-1} in (a) was also shifted to 1597 cm^{-1} in (b) and (c) thus pointing to changes in carboxyl symmetry after chromium binding. The symmetric C=O stretch was similarly shifted from 1448 cm^{-1} in (a) to 1409 cm^{-1} and 1413 cm^{-1} in (b) and (c), respectively. This decrease, coupled with a wavenumber increase of the C–O (carboxyl) stretch from 1212 cm^{-1} after binding pointed to coordination between seaweed carboxyl groups and chromium ions.

Sulphonate groups also contributed to both Cr(III) and Cr(VI) binding to *Ulva* spp., with slight differences in band shifting observed between oxidation states. Cr(III) binding resulted in a large wavenumber decrease for both the asymmetric (1336–1320 cm^{-1}) and symmetric (1163–1144 cm^{-1}) $–SO_3$ bands while Cr(VI) binding resulted in the disappearance of these bands, possibly indicating greater involvement of $–SO_3$ in Cr(VI) binding.

Biomass hydroxyl groups also played a major role in chromium binding as shown by changes in peak wavenumber and shape. Large wavenumber increases of the C–O (alcohol) band from 1006 cm^{-1} (a) to 1034 cm^{-1} (b) and (c) were observed with the appearance of an additional peak shoulder at 992 cm^{-1} in the Cr(III)-loaded sample and 984 cm^{-1} in the Cr(VI)-loaded sample.

While all seaweed–metal combinations showed significant contributions of carboxyl, amino and sulphonate groups, their relative importance varied between species and metal oxidation state. Participation of hydroxyl and ether functionalities in chromium binding also varied greatly between species. Only the green seaweeds and the red seaweed *P. lanosa* showed any interaction between chromium and biomass hydroxyl groups. On the other hand, ether participation occurred only in brown seaweeds and for Cr(VI) binding to the red seaweeds. Therefore, the effects of chromium binding to these seaweeds must be evaluated with both differences in species and metal ion in mind.

4. Conclusions

A methodology for screening various seaweeds for chromium biosorption was demonstrated in this work. Comparison of metal biosorption performance was based on expressing metal uptake against key biosorption parameters such as solution pH, equilibrium time and initial metal concentration. Chromium biosorption was highly pH dependent with higher and lower pH values favouring Cr(III) and Cr(VI) removal respectively. Cr(VI) binding also required longer exposure times than Cr(III) to ensure that equilibrium had been reached. It was shown that, at high C_i, RE values were related to the quantity of binding sites available on the seaweed. Langmuir Isotherm analysis indicated that consideration of both q_{max} and b values are necessary when determining the most suitable seaweed for chromium biosorption. FTIR analysis indicated involvement of a number of functionalities in chromium binding, the extent of which varied between seaweed species and metal oxidation state. The results obtained in this study therefore warrant further investigation into the practical applicability of dried seaweed biomass as a biosorbent for chromium-loaded waste streams.

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